

IoT Field Training

*Hardware & Embedded Systems Research Report
The ESP32 Ecosystem • Communication Protocols •
Power Electronics & Sizing*



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SECTION 1: THE ESP32 ECOSYSTEM

Espressif's ESP32 family has grown from a single Wi-Fi/Bluetooth microcontroller into a broad portfolio of System-on-Chips (SoCs) spanning two CPU architectures, multiple connectivity profiles, and price points from under \$1 to several dollars in volume. This section surveys the core variants relevant to general embedded and IoT project work, compares them side-by-side, and weighs the ESP32 family against traditional microcontrollers such as the STM32 and ATmega328P.

1.1 BOARD SURVEY

ESP32 (Classic) – WROOM / WROVER

The original ESP32 uses a dual-core Xtensa LX6 CPU running up to 240 MHz, with 802.11 b/g/n Wi-Fi and Bluetooth Classic + BLE 4.2. "WROOM" and "WROVER" are Espressif's module packaging of the same chip: WROOM modules use external SPI flash only (typically 4 MB), while WROVER modules add external SPI PSRAM (typically 8 MB), useful for camera buffers, TFT frame buffers, or larger heap needs. Neither has native USB – programming goes through an onboard USB-to-UART bridge chip (CP2102 or CH340).

ESP32-S2

A single-core Xtensa LX7 part that drops Bluetooth entirely to focus on Wi-Fi-only, cost- and power-sensitive applications. Its standout feature is native USB On-The-Go (OTG), letting it act as a USB host or device without an external bridge chip.

ESP32-S3

Dual-core Xtensa LX7 at 240 MHz, with Bluetooth (BLE 5.0) restored alongside Wi-Fi 4, plus native USB OTG and vector instructions that accelerate AI/ML inference (e.g., wake-word detection, image classification). This combination has made the S3 the flagship choice for new AIoT designs.

ESP32-C3

Espressif's first mainstream RISC-V part: single-core RV32IMC at 160 MHz, with Wi-Fi 4 and BLE 5.0. No native USB OTG, but it has a USB-Serial/JTAG peripheral for programming without a bridge chip. Known for low cost, small footprint, and hardware security features (secure boot, flash encryption).

ESP32-P4

A high-performance dual-core RISC-V part (up to 400 MHz) with an additional 40 MHz low-power core, native USB 2.0 High-Speed OTG, MIPI-CSI/DSI interfaces for cameras/displays, and hardware H.264 encoding – but no built-in Wi-Fi or Bluetooth. Designed as a companion MCU for HMI, video, and edge-vision workloads, typically paired with a smaller wireless chip (e.g., ESP32-C6).



SECTION 1: THE ESP32 ECOSYSTEM

1.2 COMPARISON TABLE

Variant	Core Architecture	CPU / Clock Speed	Wi-Fi / BLE	Native USB	Ideal Use Case
ESP32 (WROOM/WROVER)	Xtensa LX6	Dual-core, up to 240 MHz	Wi-Fi 4 (b/g/n), BT Classic + BLE 4.2	No (USB-UART bridge chip)	General-purpose IoT, legacy projects, camera/display work (WROVER + PSRAM)
ESP32-S2	Xtensa LX7	Single-core, up to 240 MHz	Wi-Fi 4 only (no Bluetooth)	Yes – USB OTG	Low-power Wi-Fi-only sensors, USB HID/host devices
ESP32-S3	Xtensa LX7	Dual-core, up to 240 MHz	Wi-Fi 4, BLE 5.0	Yes – USB OTG	AIoT flagship: voice/vision ML, smart displays, general wireless products
ESP32-C3	RISC-V (RV32IMC)	Single-core, up to 160 MHz	Wi-Fi 4, BLE 5.0	No (USB-Serial/JTAG only)	Low-cost secure IoT nodes, smart plugs, simple sensors
ESP32-P4	RISC-V (dual HP core + LP core)	Dual-core up to 400 MHz + 40 MHz LP core	None built-in (pairs with a co-processor chip for RF)	Yes – USB 2.0 High-Speed OTG	Edge-AI vision, HMI/large displays, H.264 video, robotics compute

1.3 ESP32 VS. TRADITIONAL MICROCONTROLLERS – PROS & CONS

Advantages of ESP32	Disadvantages of ESP32
Integrated Wi-Fi + Bluetooth removes need for external radio module	Not truly real-time – Wi-Fi/BLE stack interrupts and RTOS scheduling add jitter unsuited to hard real-time control loops
Far higher clock speed/RAM than ATmega328P (240 MHz / 320+ KB SRAM vs. 16 MHz / 2 KB)	Higher active and deep-sleep current than ATmega328P or low-power STM32L parts – real constraint for battery designs

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SECTION 1: THE ESP32 ECOSYSTEM

1.3 ESP32 VS. TRADITIONAL MICROCONTROLLERS – PROS & CONS

Advantages of ESP32	Disadvantages of ESP32
Low unit cost for the integrated functionality (radio + MCU + crypto in one part)	Weaker analog peripherals: non-linear/noisy ADC near rails vs. STM32's 12-bit ADCs; no true DAC on most variants
Rich free tooling: Arduino core, ESP-IDF (FreeRTOS), MicroPython, huge community	STM32 offers more precise timers, more robust HAL determinism, more industrial/automotive-certified parts
Built-in OTA firmware update support in the SDKs	Single-vendor (Espressif) supply chain vs. many second sources for STM32/AVR

Summary: Choose ESP32 for wireless connectivity and compute headroom where hard real-time isn't required. Choose STM32 for precise timing, rich analog I/O, or industrial certification. Choose ATmega328P for simple, cost/power-constrained projects with no networking need.

SECTION 2: COMMUNICATION PROTOCOLS

2.1 UART (UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER)

- **Topology:** Point-to-point only. 2 wires (TX, RX) + GND. No native addressing; can't support multiple devices on one pair.
- **Speed/Distance:** 9,600 bps-115,200 bps common (up to ~5 Mbps on some MCU hardware). Practical distance a few meters; RS-232 standard specifies ~15 m at low baud.
- **Arbitration:** None. Multiple devices need one UART per device, a mux chip, or a software addressing scheme.
- **Pros:** Simple hardware/protocol, minimal pins, universally supported, easy to debug.
- **Cons:** No multi-drop, both ends must agree on baud rate in advance, minimal error checking, noise-sensitive over distance.
- **Applications:** GPS receivers, GSM/Bluetooth serial modules, MCU debug consoles, simple sensor breakouts.

2.2 I2C (INTER-INTEGRATED CIRCUIT)

- **Topology:** Multi-controller/multi-target on a shared 2-wire bus: SDA (data) + SCL (clock) + GND. Devices identified by 7-bit (or 10-bit) address.
- **Speed/Distance:** Standard 100 kbps, Fast 400 kbps, Fast+ 1 Mbps, High-speed up to 3.4 Mbps. Distance typically under ~1 m due to bus capacitance.
- **Arbitration:** Natively multi-drop; supports multi-controller arbitration (controllers back off if another drives a dominant bit) and clock stretching (slow devices pause the bus).
- **Pros:** Only 2 wires regardless of device count; built-in addressing and ACK bit.
- **Cons:** Slower than SPI, short bus length, one misbehaving device can lock the bus, address conflicts must be managed manually.
- **Applications:** Temperature/IMU sensors, EEPROMs, RTC modules, OLED/LCD controllers, PMICs.



SECTION 2: COMMUNICATION PROTOCOLS

2.3 SPI (SERIAL PERIPHERAL INTERFACE)

- **Topology:** Single controller/multiple targets. 4 wires minimum: SCLK, MOSI, MISO, plus one dedicated CS line per device – wire count grows with device count.
- **Speed/Distance:** Commonly 1-20 MHz, up to 40-80 MHz on many MCUs. Distance is board-level (centimeters); longer cables cause signal reflection issues.
- **Arbitration:** None – controller asserts the CS line of the target it wants; only one active device at a time.
- **Pros:** Highest raw throughput of the three chip-level buses, full-duplex, low latency, simple protocol.
- **Cons:** Pin count scales poorly with device count, no built-in error checking, very short max cable length.
- **Applications:** SD/microSD cards, TFT/OLED displays, external flash/FRAM, ADCs/DACs, high-sample-rate sensors.

2.4 RS-485

- **Topology:** Multi-drop differential bus. Half-duplex: 2 wires (A/B) + GND. Full-duplex: 4 signal wires + GND. Requires a transceiver IC (e.g., MAX485) between the UART and the bus.
- **Speed/Distance:** Up to ~10 Mbps at very short range; typically 9.6 kbps–1 Mbps over long runs – up to ~1,200 m (4,000 ft) at low baud rates.
- **Arbitration:** No native arbitration – the electrical standard just defines robust differential signaling supporting up to 32 standard unit loads. Multi-device coordination (addressing, polling) is handled by a higher-layer protocol, most commonly Modbus RTU. Requires 120 Ω termination resistors at each bus end, plus bias resistors.
- **Pros:** Excellent noise immunity, very long cable runs, true multi-drop, industrial standard.
- **Cons:** Needs external transceiver chip, half-duplex needs careful turnaround timing, no native addressing, termination/biasing must be correct or the bus is unreliable.
- **Applications:** Modbus RTU industrial networks, building automation (HVAC), motor drives/VFDs, PLC-sensor networks.

2.5 I2S (INTER-IC SOUND)

- **Topology:** Synchronous serial bus purpose-built for digital audio. Typically point-to-point (one controller driving one or more receivers). 3-4 wires: BCLK (bit clock), WS/LRCLK (word select – left/right channel), and SD (serial data; separate SDIN/SDOUT lines for full duplex) + GND. Not a general multi-drop bus – usually one dedicated device per bus.
- **Speed/Distance:** Bit clock rate = sample rate $\times$ bit depth $\times$ channels (e.g., 44.1 kHz stereo/16-bit $\approx$ 1.4 MHz BCLK). Distance is board-level to short cable runs (centimeters up to ~1 m with clean routing).
- **Arbitration:** None – it's a dedicated synchronous stream, not a shared/addressed bus. The controller drives BCLK/WS continuously; there's no contention to resolve.
- **Pros:** Purpose-built for sample-accurate digital audio, low CPU overhead via DMA streaming, supports simultaneous mic-in/speaker-out (full duplex).
- **Cons:** Not general-purpose, no addressing, effectively one (or few) dedicated audio devices per bus, short range, requires careful clock/timing design.
- **Applications:** I2S MEMS microphones, DAC/audio codec chips, digital amplifiers (e.g., MAX98357), voice-assistant and voice-command embedded projects.



SECTION 2: COMMUNICATION PROTOCOLS

2.6 CAN BUS (CONTROLLER AREA NETWORK)

- **Topology:** Multi-master, multi-node differential bus. 2 wires (CAN_H, CAN_L, twisted pair) + common ground reference. Every node sees every message – messages carry a priority/identifier rather than a destination address (broadcast bus).
- **Speed/Distance:** Classical CAN: up to 1 Mbps at ≤40 m, trading speed for distance (e.g., 125 kbps to ~500 m, 50 kbps to ~1 km). CAN FD (Flexible Data-Rate) pushes the data phase up to 5-8 Mbps.
- **Arbitration:** Non-destructive bitwise arbitration – nodes transmit dominant/recessive bits on the identifier field, and the lowest identifier value automatically wins without any data loss or retransmission (CSMA/CR). The highest-priority message always gets through first.
- **Pros:** Extremely robust in electrically noisy environments (motor drives, automotive); true multi-master with no central controller; built-in error detection/correction (CRC, ACK, bit-stuffing); guaranteed message prioritization.
- **Cons:** Needs a dedicated CAN transceiver IC (e.g., MCP2551, TJA1050) – the ESP32's built-in TWAI controller still needs an external transceiver; small payload per frame (8 bytes classical CAN, up to 64 bytes CAN FD); requires 120 Ω termination at each bus end; more complex protocol stack than UART/I2C.
- **Applications:** Robotics (joint/motor controller networks, e.g., ODrive or ROS2 CAN nodes), automotive ECUs, industrial automation, drone flight controllers, EV battery management systems (BMS).

2.7 PROTOCOL SUMMARY COMPARISON

Protocol	Wire Count	Max Practical Distance	Typical Speed	Multi-Device Support
UART	2 + GND	~15 m (low baud)	9.6 kbps–5 Mbps	None (point-to-point only)
I2C	2 + GND (shared)	~1 m	100 kbps–3.4 Mbps	Yes – native 7/10-bit addressing
SPI	3 + 1 per device	Centimeters (board-level)	1–80 MHz	Yes – via dedicated CS lines
RS-485	2 (half)/4 (full) + GND	Up to 1,200 m	9.6 kbps–1 Mbps (long-run typical)	Yes – up to 32 nodes, via higher-layer protocol
I2S	3–4 + GND	Centimeters–1 m	Audio-rate (~1–6+ MHz bit clock)	No – dedicated point-to-point stream
CAN Bus	2 + GND	Up to 1 km (at lower baud)	125 kbps–1 Mbps (up to 8 Mbps CAN FD)	Yes – broadcast bus, priority-based arbitration



SECTION 3: POWER ELECTRONICS & SIZING FUNDAMENTALS

3.1 CONVERTERS VS. INVERTERS

AC/DC Rectifiers

Convert AC to DC using a diode bridge (typically 4 diodes) that forces current in one direction, producing pulsating DC. A bulk filter capacitor smooths this into rough DC, and a linear/switching regulator holds a fixed output.

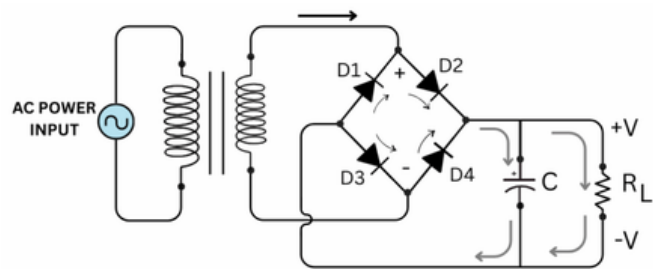
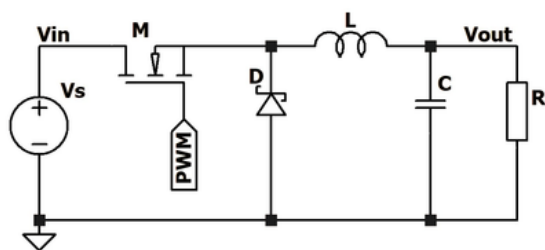


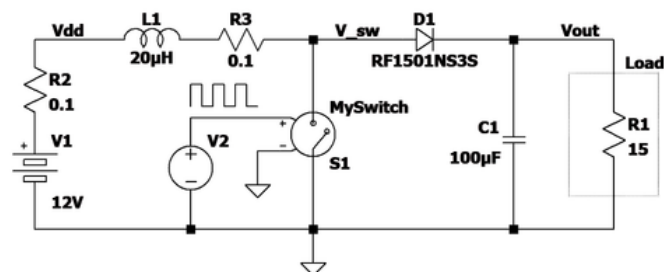
Figure 1: Full Bridge Rectifier Circuit

DC/DC Converters:

- **Buck (step-down):** Takes a higher DC input and produces a lower regulated output. A switching transistor rapidly chops the input (PWM); an inductor + output capacitor average this into a smooth lower DC voltage. Efficiency typically 85-95%. Example: 12 V battery → 5 V/3.3 V logic.
- **Boost (step-up):** Takes a lower DC input and produces a higher output. While the switch is closed, the inductor stores energy; when it opens, that energy releases through a diode into the output capacitor, pushing output above input. Example: 3.7 V lithium cell → 5 V USB output.



(2.1)



(2.2)

Figure 2: Figure A shows a Buck converter circuit, and Figure B shows a Boost converter circuit.

DC/AC Inverters

Convert DC to AC using switches (H-bridge or push-pull, MOSFETs/IGBTs) that alternately reverse polarity across the load in a controlled pattern, synthesizing an AC waveform. Simple designs produce square/modified-sine waves; better ones use high-frequency PWM + LC filtering for a smooth sine wave. Basis of solar inverters, UPS backup power, and Variable Frequency Drives (VFDs).



SECTION 3: POWER ELECTRONICS & SIZING FUNDAMENTALS

3.1 CONVERTERS VS. INVERTERS

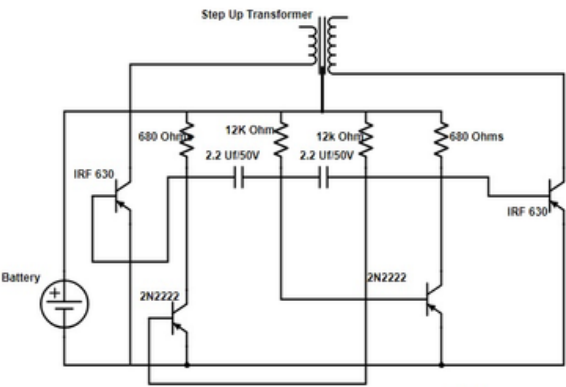


Figure 3: DC to AC Power Converter circuit

Key Distinction
Rectifier: AC → DC (diode bridge + filter).
DC/DC Converter: DC → DC at a different voltage (switch + inductor, buck steps down, boost steps up).
Inverter: DC → AC (switch bridge synthesizes an alternating waveform).

3.2 POWER SUPPLY SIZING GUIDE – WORKED EXAMPLE (PROJECT 2)

Step 1 – List every load and its rated current

Component	Supply Rail	Typical Current	Notes
DC motor	5 V	80 - 150 mA	Using worst case: 150 mA.
Relay module	5 V (logic side)	70 mA	Coil + driver transistor + status LED
Current sensor	5 V	10 mA	Quiescent supply current
ESP32 dev board	5 V	30 - 70 mA	using 70 mA to stay safe
4× status LEDs	3.3 V/5 V	20 mA each	20 mA each
stop push button	3.3 V	< 1 mA	Internal pull resistor only

Note on the stall current estimate of the DC motor: Stall current not specified on datasheet, small brushed gear motors typically stall at 5-8× rated running current.

Step 2 – loads addition

Since everything is 5V-class, this collapses to one rail:
Motor (150 mA) + relay (70 mA) + sensor (10 mA) + ESP32 (70 mA) + LEDs (80 mA) = 380 mA continuous



3.2 POWER SUPPLY SIZING GUIDE – WORKED EXAMPLE (PROJECT 2)

Good practice is to size a supply for 125–150% of the calculated continuous load, so the supply runs comfortably below its maximum rating (lower heat, longer life, and headroom for measurement tolerance).

$380 \text{ mA} \times 1.5 \approx 570 \text{ mA}$ minimum continuous rating

Worst realistic case is the motor stalling:

1,000–1,200 mA (motor stall) + 230 mA (everything else) = 1.23–1.43 A momentary peak
(This is brief – the current sensor should trip the relay within milliseconds of detecting the stall, so the supply only needs to survive this peak momentarily, not continuously.)

570 mA continuous $\times$ 5V $\approx$ 2.8 W continuous budget; 1.43 A peak $\times$ 5V $\approx$ 7.15 W momentary.

Option	Rating	Notes
Recommended	a 5V, 1.5A (7.5W) Regulated supply	Comfortable headroom above both the 570 mA continuous need and the 1.43 A stall scenario
Minimum acceptable	5V, 1A	Only safe if you're confident the motor won't stall for more than a few ms before the relay trips.

Figure 4: Project 2 schematic

CONCLUSION & WHAT I LEARNED & REFERENCES

WHAT I LEARNED

ESP32 ecosystem: "More powerful" isn't a universal metric. The real skill is mapping a project's actual requirements onto each variant's trade-off space — architecture, connectivity, and I/O aren't independent choices, they're one design decision.

Protocols: UART, I2C, SPI, RS-485, I2S, and CAN all trade off the same three things against each other — wire count, speed, and distance — and none wins on all three. Protocol selection isn't memorization, it's identifying which trade-off a given application actually needs.

Power sizing: This is where the process got tested against a real, imperfect component. The motor's datasheet gave running current but not stall current, forcing an estimate rather than a hard number — a reminder that real datasheets are often incomplete, and that gaps need to be flagged, not guessed silently.

CONCLUSION

This project reinforced that component selection is rarely a matter of matching a spec sheet to a requirement — it's an iterative process of refining assumptions against real constraints.

Overall takeaway: the most transferable lesson from this project wasn't any single fact about the ESP32, a specific protocol, or a converter topology — it was the discipline of separating what a component's datasheet actually says from what I was assuming it said, and separating what the design should include from what merely felt safe to include. That distinction is what separates a design that's genuinely sized correctly from one that's simply padded with untraceable margin.

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